

6-4c Weighted Average Cost Method

The weighted average cost method uses the weighted average unit cost for determining the cost of goods sold and the ending inventory. If purchases are relatively uniform during a period, the weighted average cost method provides results that are similar to the physical flow of goods.

The weighted average unit cost is determined as follows:

$$\text{Weighted Average Unit Cost} = \frac{\text{Total Cost of Units Available for Sale}}{\text{Units Available for Sale}}$$

To illustrate, the data for **Item 127B** are used as follows:

$$\begin{aligned} \text{Weighted Average Unit Cost} &= \frac{\text{Total Cost of Units Available for Sale}}{\text{Units Available for Sale}} = \frac{\$45,180}{2,100 \text{ units}} \\ &= \$21.51 \text{ per unit (rounded)} \end{aligned}$$

The cost of the January 31 ending inventory is as follows:

$$\text{Inventory, January 31: } \$17,208 \text{ (800 units} \times \$21.51)$$

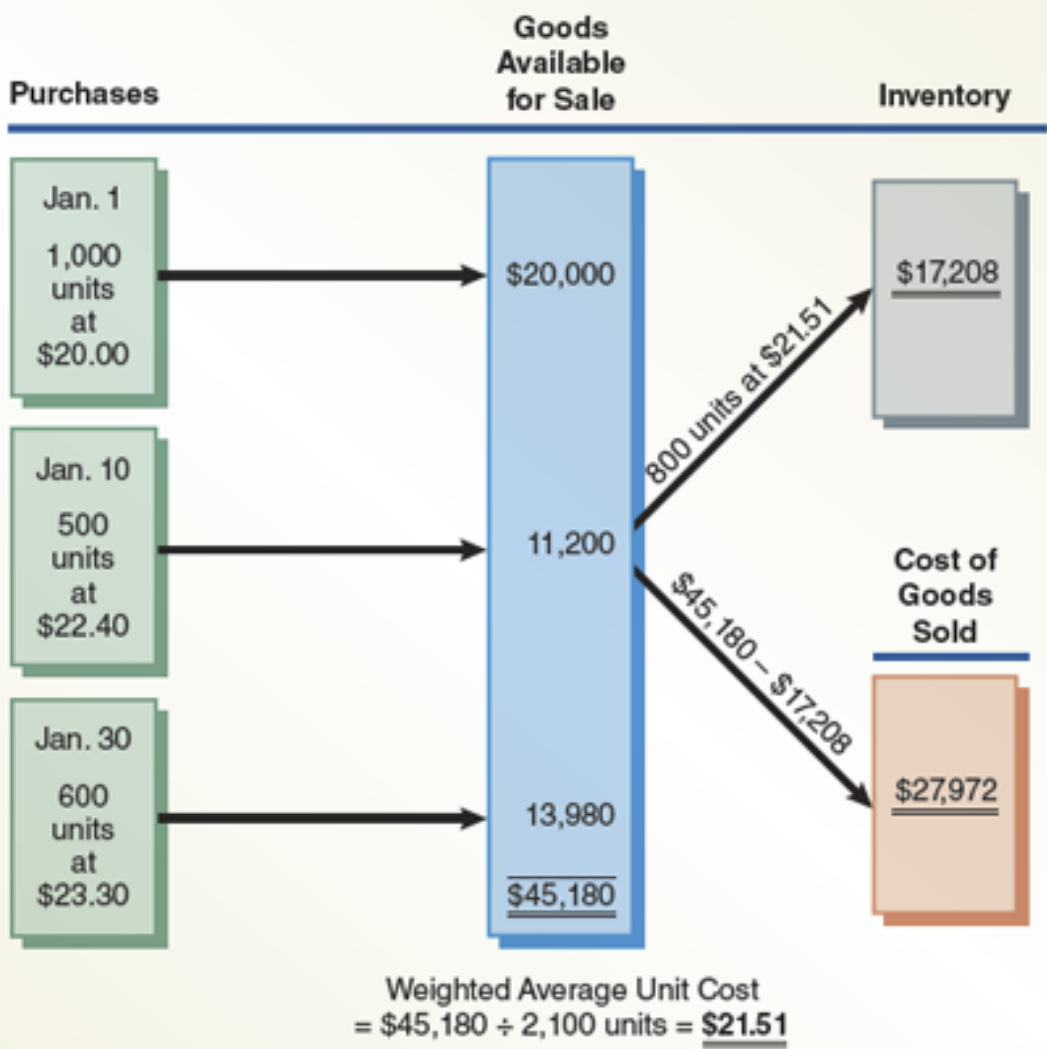
Deducting the cost of the January 31 inventory of \$17,208 from the cost of goods available for sale of \$45,180 yields the cost of goods sold of \$27,972, computed as follows:

Beginning inventory, January 1	\$ 20,000
Purchases (\$11,200 + \$13,980)	<u>25,180</u>
Cost of goods available for sale in January	\$ 45,180
Ending inventory, January 31	<u>(17,208)</u>
Cost of goods sold	<u><u>\$ 27,972</u></u>

The \$17,208 cost of the ending inventory on January 31 is made up of a weighted average of the cost of units available for sale during the month. The cost of goods sold is made up of the weighted average of the cost of the units sold during the month. **Exhibit 8** shows the relationship of the cost of goods sold for January and the ending inventory on January 31.

Exhibit 8

Weighted Average Flow of Costs with Periodic Inventory



Check up Corner 6-3

Periodic Inventory

The beginning inventory, purchases, and sales of Item PEAR4 for a recent year are as follows:

Jan. 1	Inventory	6 units at \$50	\$ 300
Mar. 20	Purchase	14 units at \$55	770
Oct. 30	Purchase	20 units at \$62	1,240
	Available for sale	40 units	<u>\$2,310</u>

There are 16 units of the item in the physical inventory at December 31, the end of the fiscal year. The company uses the periodic inventory system. Determine (1) the December 31 inventory balance and (2) the cost of goods sold for the year, using the:

- a. first-in, first-out (FIFO) method.
- b. last-in, first-out (LIFO) method.
- c. weighted average cost method.

Chapter 6: Inventories: 6-4c Weighted Average Cost Method

Book Title: Financial and Managerial Accounting

Printed By: Raydel Castro Guerra (raydel.castro001@mymdc.net)

© 2023 Cengage Learning, Cengage Learning

© 2026 Cengage Learning Inc. All rights reserved. No part of this work may be reproduced or used in any form or by any means - graphic, electronic, or mechanical, or in any other manner - without the written permission of the copyright holder.